

R3

Basic Questions of if-else & for loops

Join Discord - <https://tinyurl.com/ly-discord>

Loops Question

Question 1 - Program to check whether a number is Armstrong number or not

Example

Input

371

Output

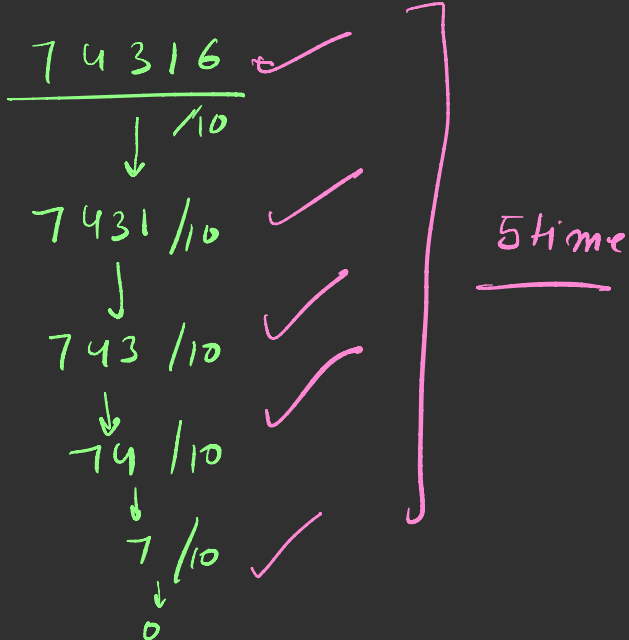
371 is Armstrong number.

$$\underline{3^3 + 7^3 + 1^3 = 371}$$

$$\underbrace{4^3 + 0^3 + 7^3}_{\text{no. of dig} = 3} = \underline{407}$$

$$407 \\ 4^3 + 0^3 + 7^3$$

Explanation slide



L4

Intermediate loops

Practice Question [Recap]

Print "FizzBuzz" if i is divisible by 3 and 5.

Print "Fizz" if i is divisible by 3.

Print "Buzz" if I is divisible by 5.

Print i if none of the above conditions are true.

1 2 Fizz 4 Buzz Fizz 7 8 Fizz
Buzz 11 Fizz 13 14 Fizz Buzz

Break and continue

Let's say in fizzbuzz, an extra condition is that for multiples of 4, do nothing. What can be done in this scenario ?

Also if there is a danger number also given, how can we stop the loop at that time ?

Loops Continued

Count the number of digits of a positive number.

Loops Continued

A sum of digits of a positive number.

$$\begin{array}{rcl} 4312 \% 10 & = & 2 \\ \hline & / & 10 \\ 431 \% 10 & = & 1 \\ & / & 10 \\ 43 \% 10 & = & 3 \\ & / & 10 \\ 4 \% 10 & = & 4 \\ 416 & \Rightarrow & \end{array}$$

Loops Continued

Sum of digits of all numbers from 1 to N.

23

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	2	3	4	5	6	7	8	9	1	2	3	4	5	6

16	17	18	19	20	21	22	23
1	8	9	6	2	3	4	5

Loops Continued

For without initialization / condition / updation

Example of printing even numbers from 1 to N using a for loop, but without doing initialization / condition / updation in the loop.

While Loop Question

Get First N numbers that have sum of digits = k

3

7

7 16 25

While Loop Question

Check whether a number is palindrome or not

maam

eye

rada

racecar

naman

nitin

Arosa

Anna

Bob

Hamam

Oppo Kulak

jhaj Nevan

Dasas

Kanak

Amma

Nayan

1 4 2 2 4 1

$$rev = 0$$

1 4 6 4 1

$$rev = 1$$

$$rev \times 10 + 4$$

1 4 6 2 3 1

$$rev = \underline{14}$$

$$rev = \underline{142}$$

Practice and Homework Questions

- Check if a given number is a power of 2 or not
 - <https://leetcode.com/problems/power-of-two/>
- Check if a given number is prime or not?
- Print all prime numbers from 1 to N.
- Calculate product of all digits
- Print all factors of a number
- Print first and last digit of a number

Thank You!

Please practice more questions and examples as above !!

Loops Question

Question 2 - Program to find Armstrong numbers between 1 to n

Example

Input

1

1000

Output

1 2 3 4 5 6 7 8 9 370 371 407

Explanation slide

Loops Question

Question 3 - Program to find prime factors of a number

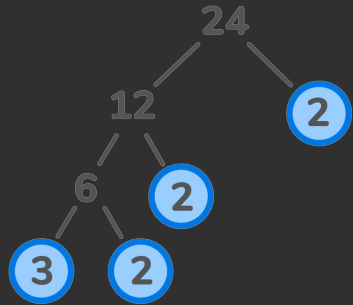
Example

Input

10

Output

2 5



Explanation slide

Loops Question

Question 4 - Print fibonacci series upto n terms

Example

Input

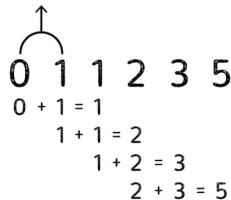
10

Output

0 1 1 2 3 5 8 13 21 34

Fibonacci Series

Default



Explanation slide

Loops Question

Question 5 - Convert number from Binary to Decimal

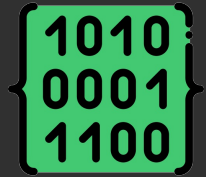
Example

Input

101

Output

5



Explanation slide

Thank You!

Please practice more questions and examples as above !!